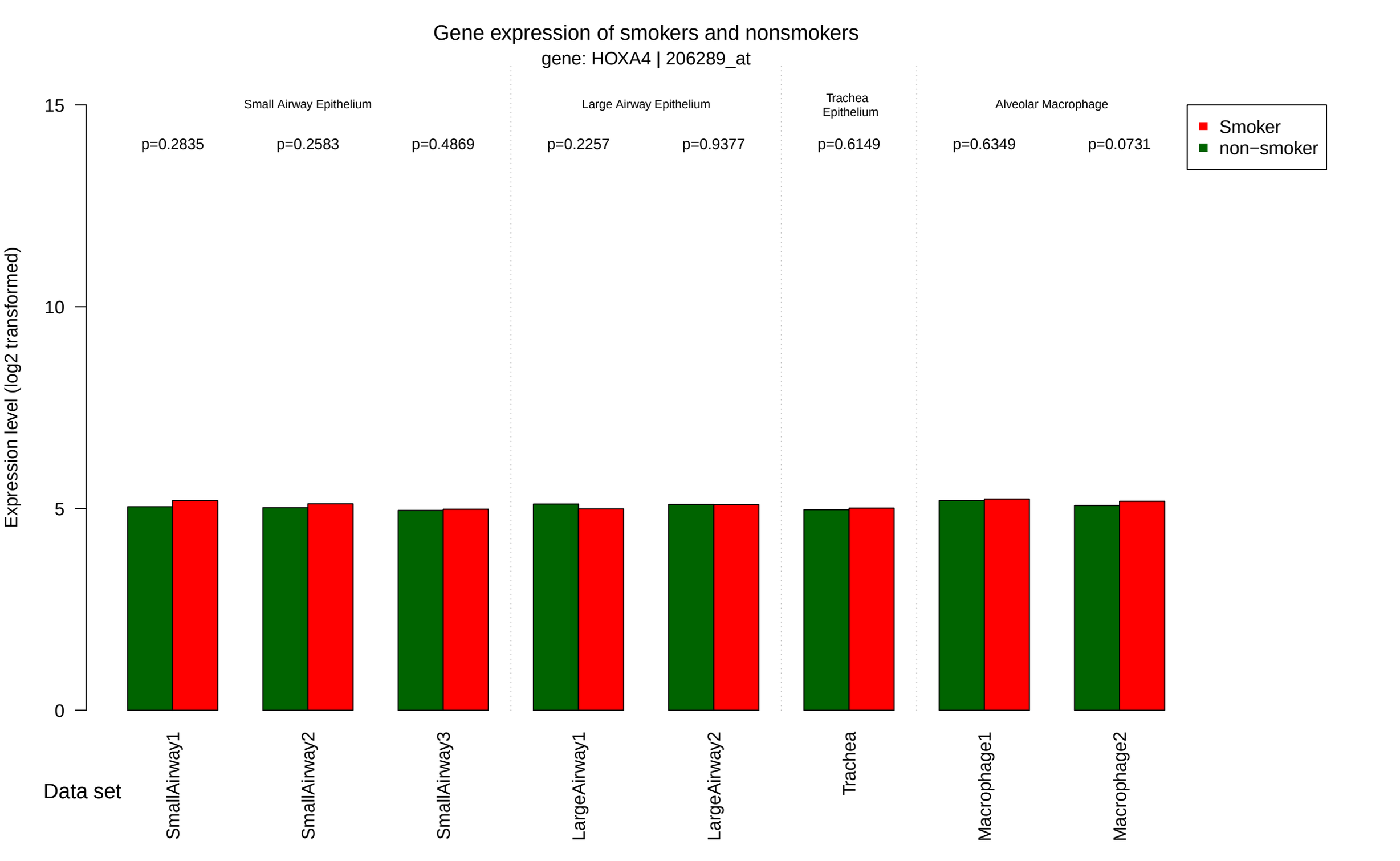
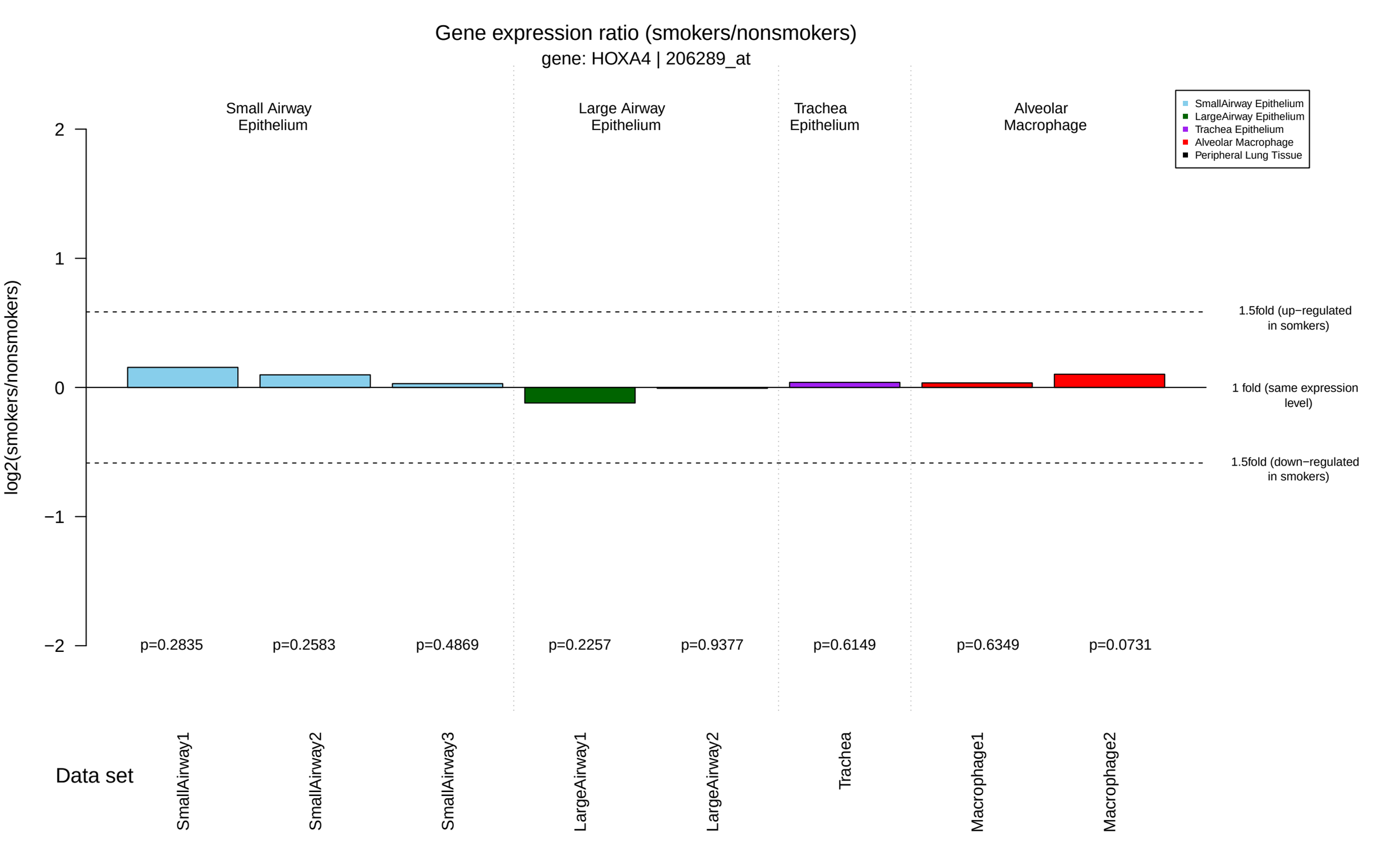
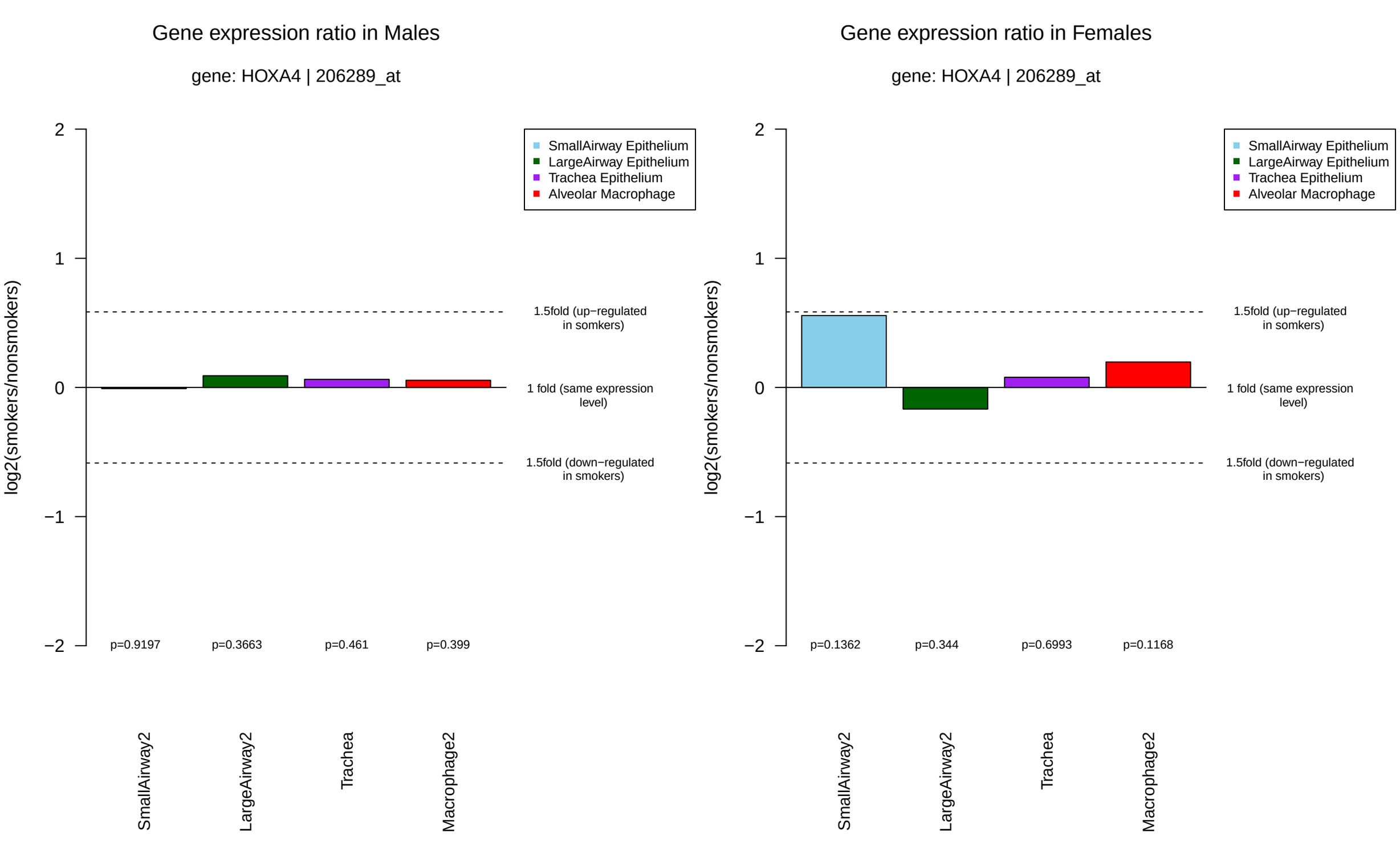
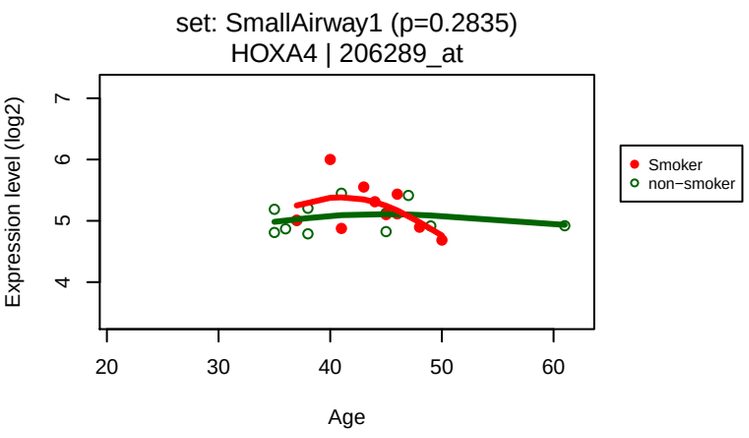
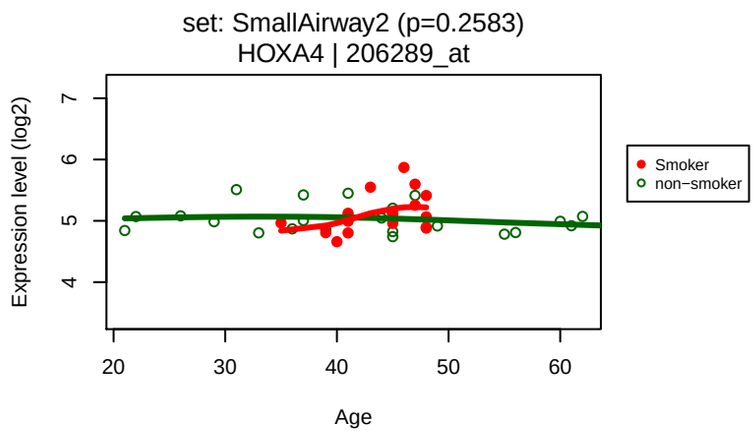
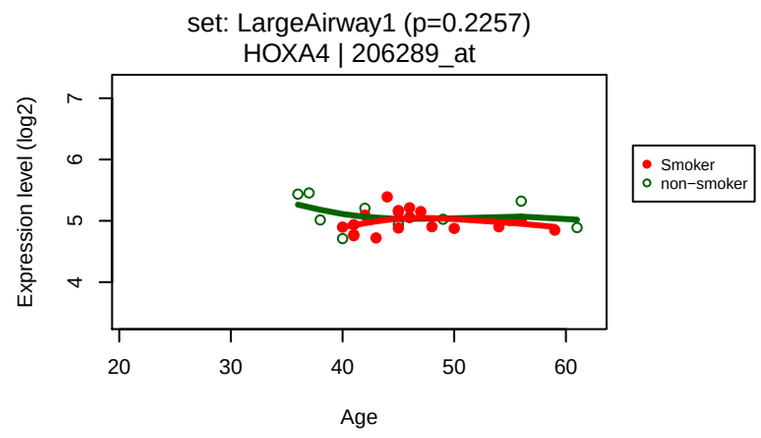

Gene expression of smokers and nonsmokers
gene: HOXA4 | 206289_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2835
p=0.2583
p=0.4869
p=0.2257
p=0.9377
p=0.6149
p=0.6349
p=0.0731
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

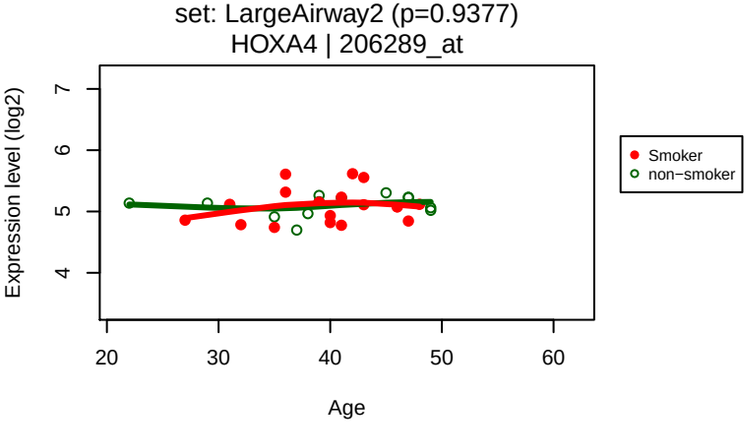

Gene expression ratio (smokers/nonsmokers)
gene: HOXA4 | 206289_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2835
p=0.2583
p=0.4869
p=0.2257
p=0.9377
p=0.6149
p=0.6349
p=0.0731
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

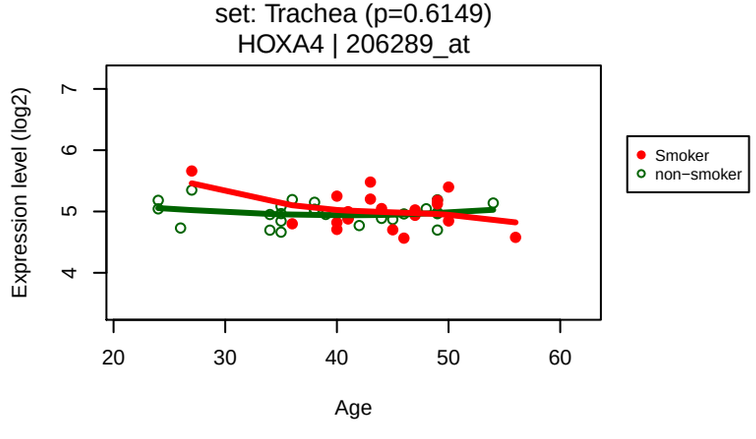

Gene expression ratio in Males
gene: HOXA4 | 206289_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9197
p=0.3663
p=0.461
p=0.399
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXA4 | 206289_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1362
p=0.344
p=0.6993
p=0.1168
SmallAirway2
LargeAirway2
Trachea
Macrophage2

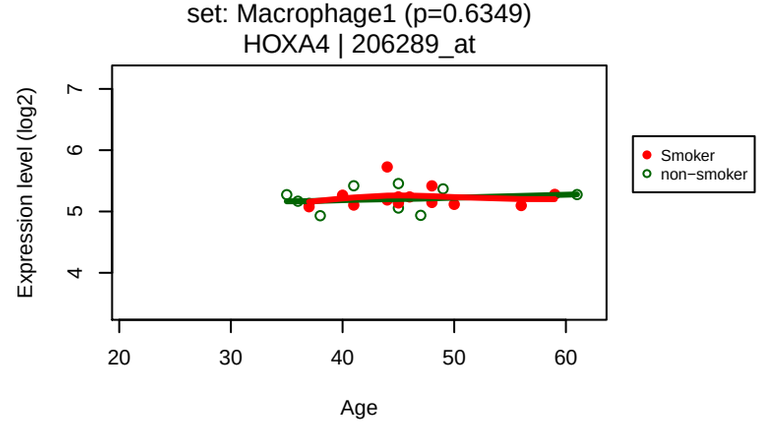

set: SmallAirway1 (p=0.2835)
HOXA4 | 206289_at
Expression level (log2)
Age
Smoker
non−smoker

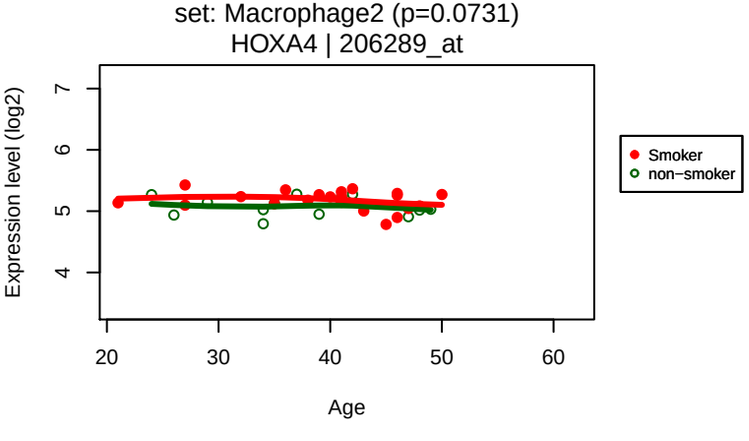

set: SmallAirway2 (p=0.2583)
HOXA4 | 206289_at
Expression level (log2)
Age
Smoker
non−smoker

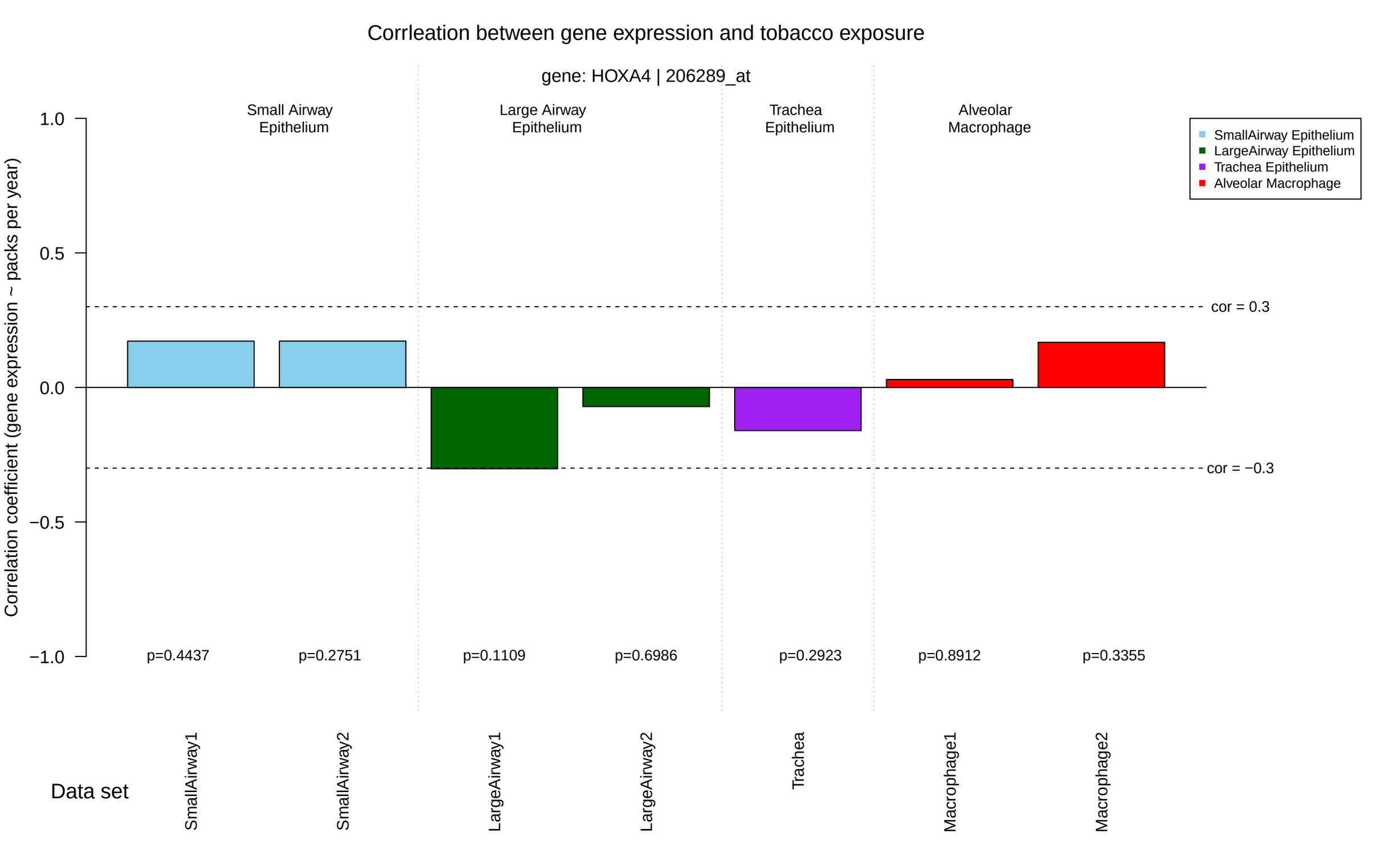

set: LargeAirway1 (p=0.2257)
HOXA4 | 206289_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9377)
HOXA4 | 206289_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6149)
HOXA4 | 206289_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6349)
HOXA4 | 206289_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0731)
HOXA4 | 206289_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXA4 | 206289_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4437
p=0.2751
p=0.1109
p=0.6986
p=0.2923
p=0.8912
p=0.3355
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2